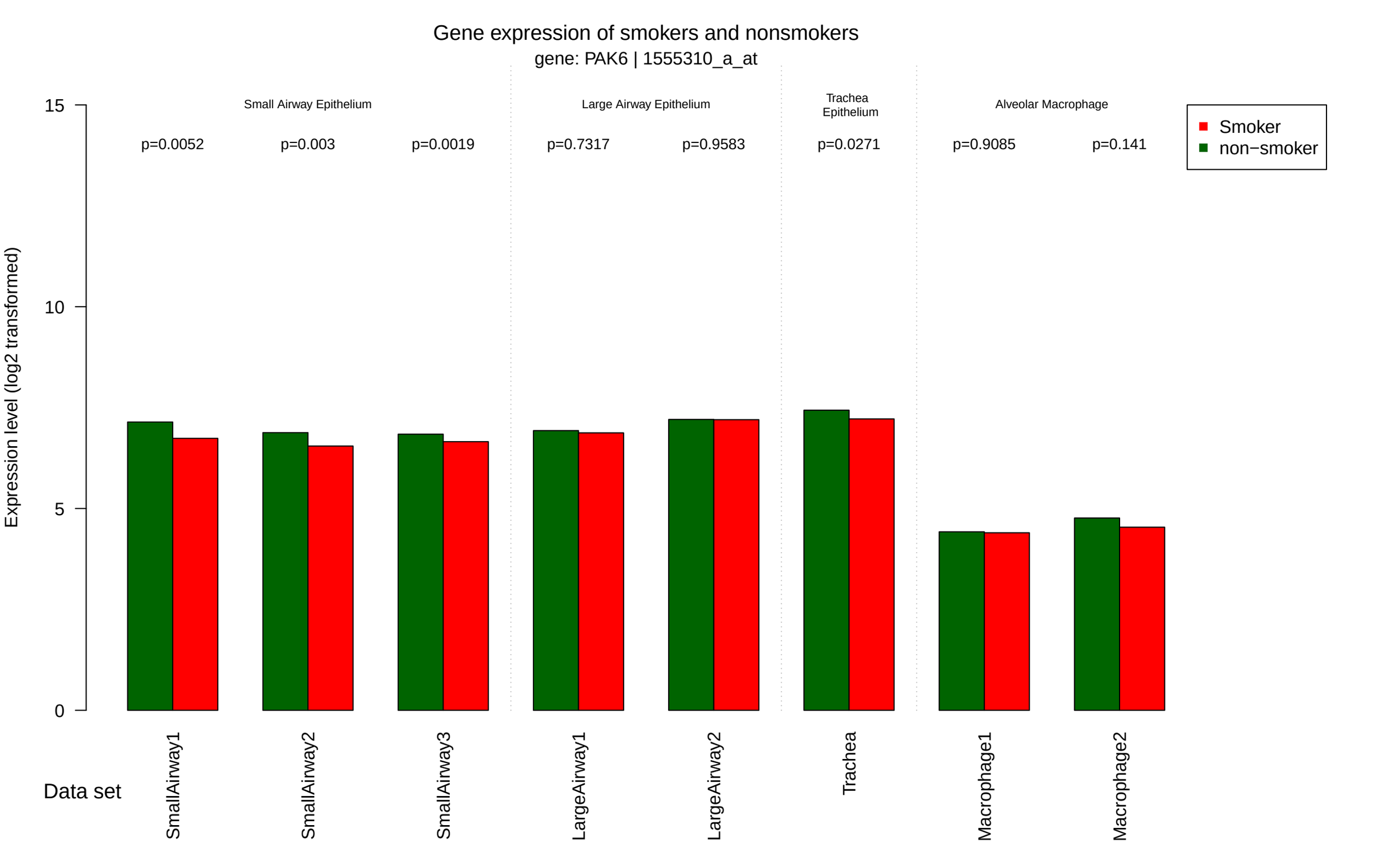
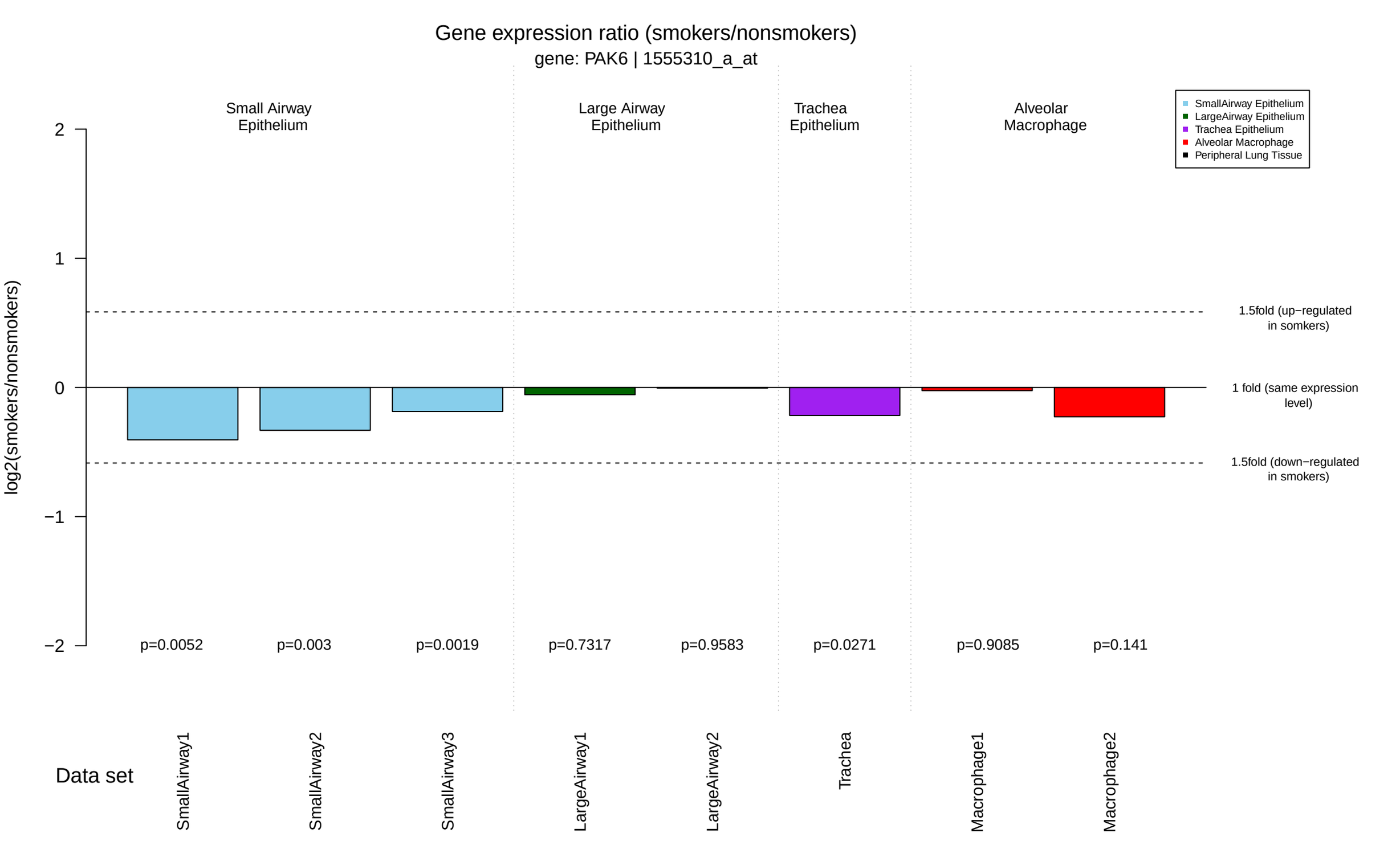
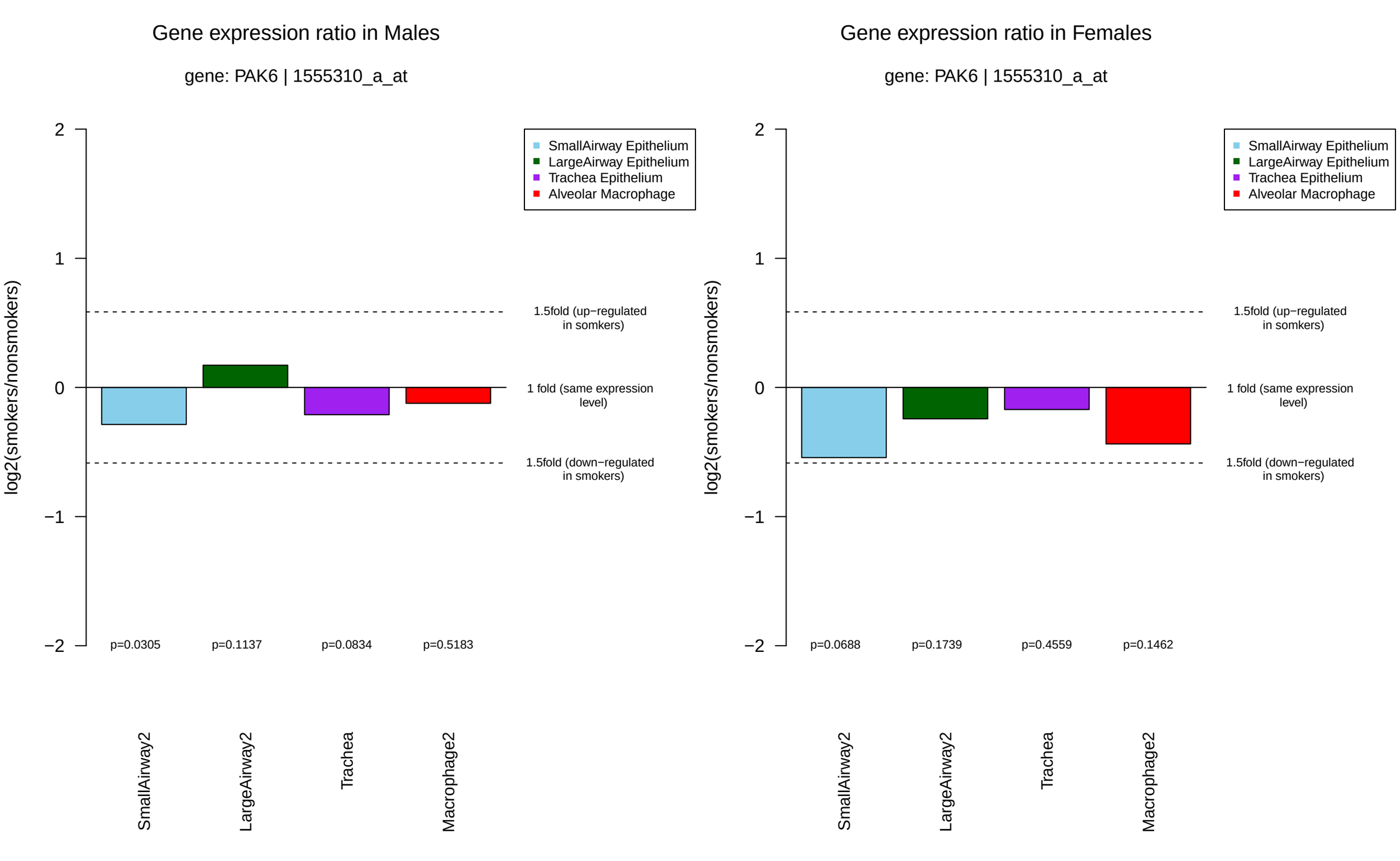
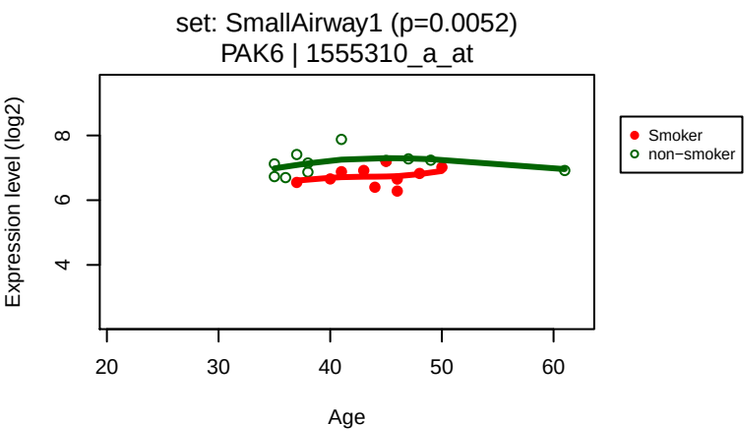
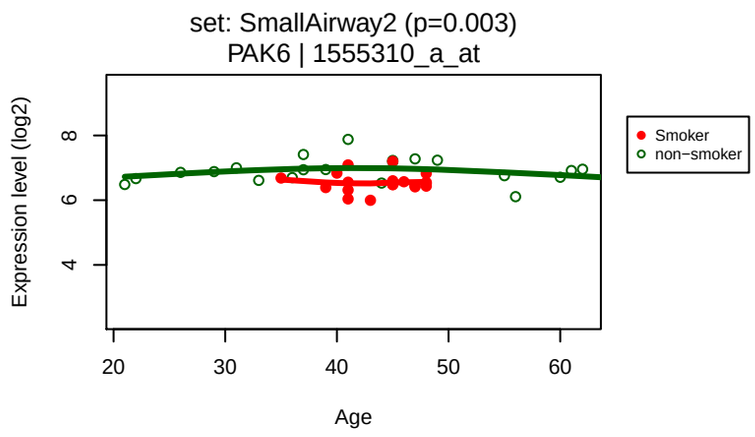
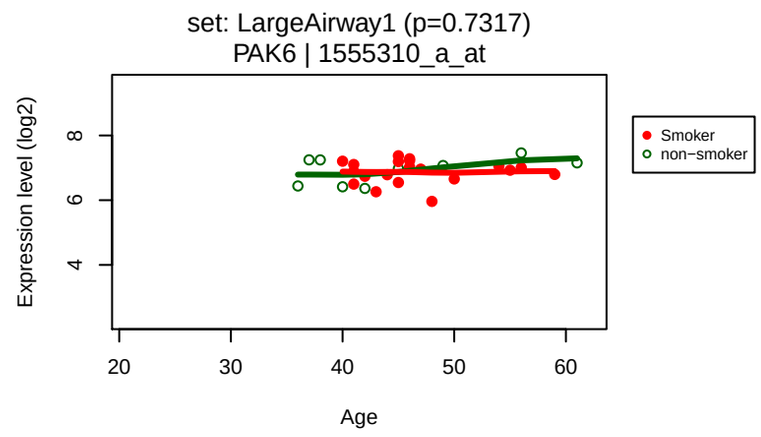

Gene expression of smokers and nonsmokers
gene: PAK6 | 1555310_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0052
p=0.003
p=0.0019
p=0.7317
p=0.9583
p=0.0271
p=0.9085
p=0.141
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

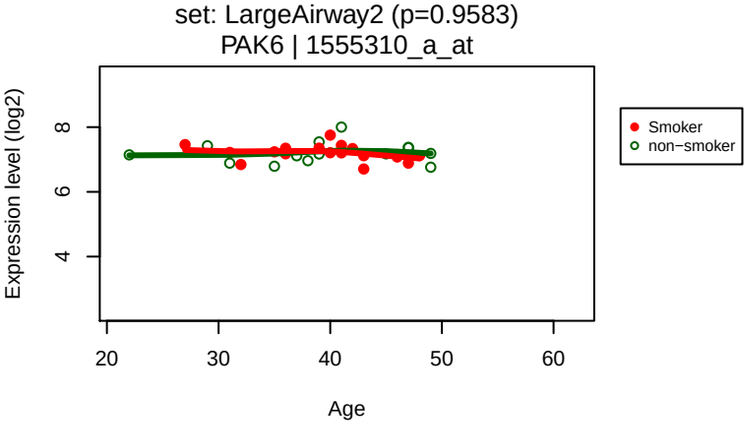

Gene expression ratio (smokers/nonsmokers)
gene: PAK6 | 1555310_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0052
p=0.003
p=0.0019
p=0.7317
p=0.9583
p=0.0271
p=0.9085
p=0.141
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

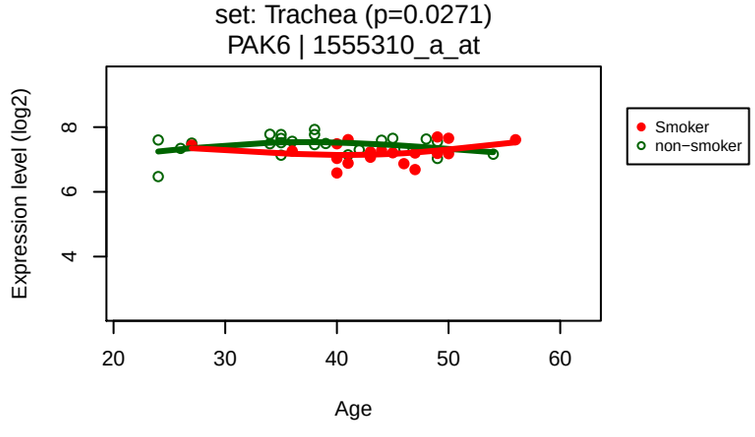

Gene expression ratio in Males
gene: PAK6 | 1555310_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0305
p=0.1137
p=0.0834
p=0.5183
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAK6 | 1555310_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0688
p=0.1739
p=0.4559
p=0.1462
SmallAirway2
LargeAirway2
Trachea
Macrophage2

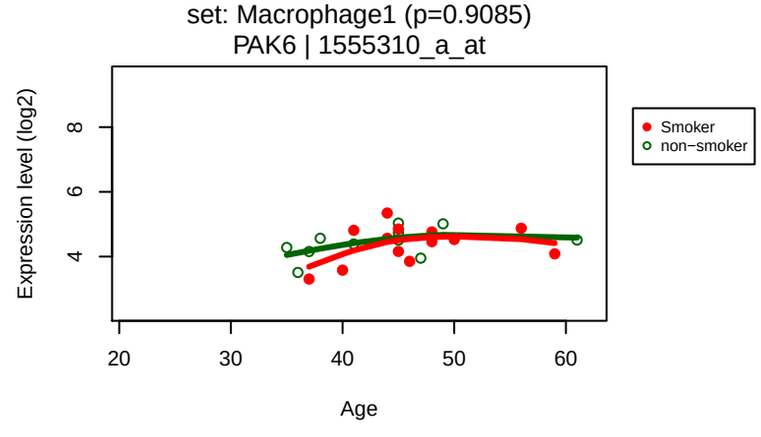

set: SmallAirway1 (p=0.0052)
PAK6 | 1555310_a_at
Expression level (log2)
Age
Smoker
non−smoker

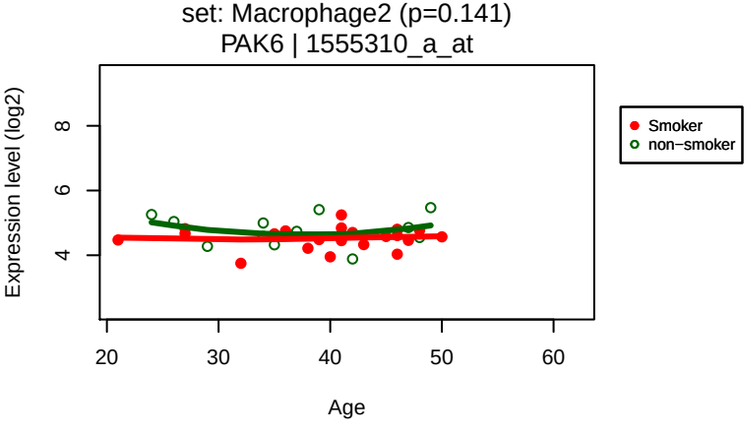

set: SmallAirway2 (p=0.003)
PAK6 | 1555310_a_at
Expression level (log2)
Age
Smoker
non−smoker

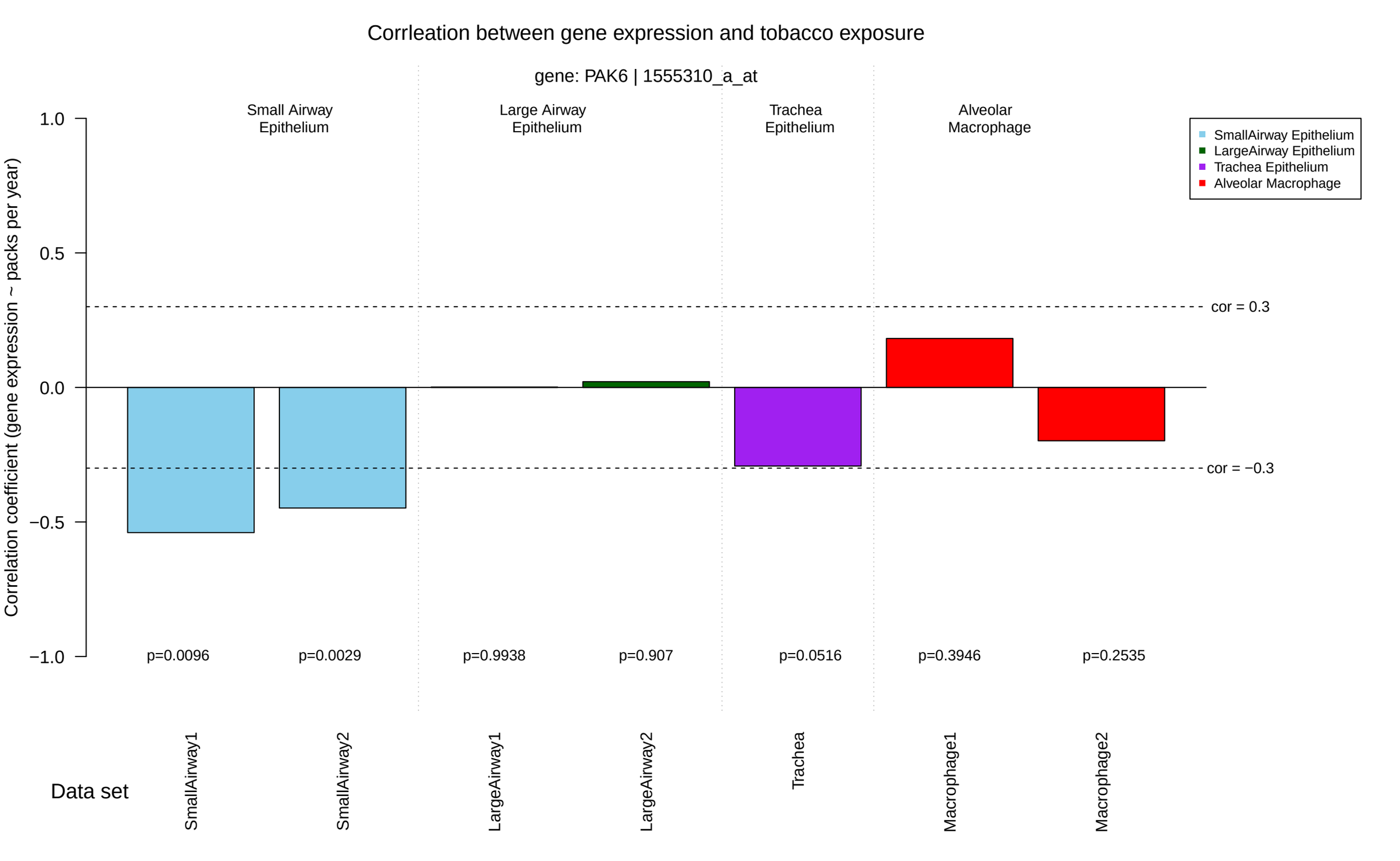

set: LargeAirway1 (p=0.7317)
PAK6 | 1555310_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9583)
PAK6 | 1555310_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0271)
PAK6 | 1555310_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9085)
PAK6 | 1555310_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.141)
PAK6 | 1555310_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAK6 | 1555310_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0096
p=0.0029
p=0.9938
p=0.907
p=0.0516
p=0.3946
p=0.2535
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2